STATE SPACE MODELS ARE EFFECTIVE SIGN LANGUAGE LEARNERS: EXPLOITING PHONOLOGICAL COMPOSITIONALITY FOR VOCABULARY-SCALE RECOGNITION

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Paper under double-blind review

ABSTRACT

Sign language recognition suffers from catastrophic scaling failure: models achieving high accuracy on small vocabularies collapse at realistic sizes. Existing architectures treat signs as atomic visual patterns, learning flat representations that cannot exploit the compositional structure of sign languages—systematically organized from discrete phonological parameters (handshape, location, movement, orientation) reused across the vocabulary. We introduce PHONSSM, enforcing phonological decomposition through anatomically-grounded graph attention, explicit factorization into orthogonal subspaces, and prototypical classification enabling few-shot transfer. Using skeleton data alone, PHONSSM achieves 72.1% on WLASL2000 (+18.4pp over skeleton SOTA) and 53.3% on a new 5,565-sign dataset, with +225% few-shot improvement.

1 INTRODUCTION

The vocabulary scaling problem. A persistent puzzle in recognition systems: models achieving near-perfect accuracy on small vocabularies ($<$100 classes) degrade catastrophically at realistic scales ($>$1,000 classes). This is not merely a data problem—performance collapses even with abundant training examples. We argue this reflects a fundamental *compositional bottleneck* in representation learning.

Consider the contrast between two strategies. *Flat representations* assign each category an independent embedding; capacity scales as $O(K)$ with vocabulary size K. *Compositional representations* factor categories into combinations of shared primitives; if K categories arise from $M \ll K$ primitives, capacity scales as $O(M)$ while covering $O(M^c)$ combinations. This exponential gap explains why humans effortlessly generalize to novel words/signs sharing familiar components, while neural networks struggle.

Sign language as a compositional testbed. Sign languages exemplify this structure. Just as spoken words decompose into phonemes, signs decompose into *cheremes*—minimal contrastive units including handshape ($\sim$30 categories), location ($\sim$15), movement ($\sim$10), and orientation ($\sim$8) (Stokoe, 1960; Battison, 1978). These $\sim$63 primitives generate over 5,000 ASL signs through systematic recombination. The sign for "mother" differs from "father" only in location; "chair" differs from "sit" primarily in movement.

Why current approaches fail. Standard architectures (LSTMs (**?**), Transformers (**?**), GCNs (**?**)) learn implicit flat representations. A Transformer may distinguish "mother" from "father," but nothing ensures it has learned the location contrast that generalizes to other minimal pairs. The signature: poor few-shot performance and non-compositional errors.

Our approach. We introduce PHONSSM (Figure 1), an architecture with explicit phonological structure: separate pathways for handshape, location, movement, and orientation with factorization objectives. We use skeleton input (MediaPipe landmarks (Lugaresi et al., 2019)) for privacy, efficiency, and domain invariance.

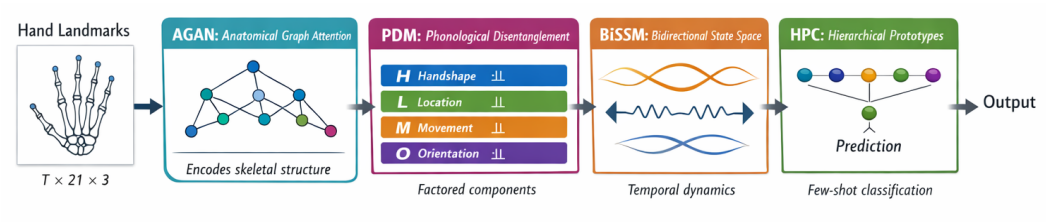

Figure 1: **PhonSSM architecture.** Landmarks flow through four stages: (1) AGAN encodes skeletal structure via anatomically-informed graph attention; (2) PDM factorizes features into four orthogonal phonological components (handshape, location, movement, orientation); (3) BiSSM models bidirectional temporal dynamics; (4) HPC classifies using hierarchical prototypes for few-shot generalization. Total: 3.2M parameters.

Contributions. (1) We formalize the *compositional bottleneck*: flat representations have $O(K)$ capacity while compositional domains have $O(M^c)$ structure. (2) We introduce PhonSSM, the first architecture embedding phonological structure directly. (3) State-of-the-art: 88.4% WLASL100, **72.1% WLASL2000** (+18.4pp over SOTA), 53.3% on 5,565 signs. (4) Causal evidence for compositionality: intervening on component embeddings flips predictions to minimal pairs 73.2% of the time.

2 Method

PhonSSM processes landmarks $\mathbf{X} \in \mathbb{R}^{T \times N \times 3}$ (T=30 frames, N=21 or 75 landmarks) through four stages. Full details are in Appendix B.

Stage 1: Anatomical Graph Attention (AGAN). Hand landmarks form a graph with anatomically-informed connectivity (finger chains, palm connections). We apply multi-head graph attention (Veličković et al., 2018) constrained to skeletal neighbors, then mean-pool over nodes to obtain per-frame spatial features $\mathbf{z}_t \in \mathbb{R}^D$.

Stage 2: Phonological Factorization (PDM). Four parallel MLPs project spatial features into orthogonal component subspaces: $\mathbf{c}_t^{(i)} = \text{MLP}_i(\mathbf{z}_t) \in \mathbb{R}^{D_c}$ for $i \in \{\text{hand}, \text{loc}, \text{mov}, \text{ori}\}$. Movement receives additional temporal convolution. An orthogonality loss $\mathcal{L}_{\text{ortho}} = \sum_{i \neq j} \cos^2(\bar{\mathbf{c}}^{(i)}, \bar{\mathbf{c}}^{(j)})$ encourages decorrelation.

Stage 3: Bidirectional SSM (BiSSM). We adapt Mamba (Gu & Dao, 2023) for bidirectional temporal modeling, running forward and backward SSMs in parallel. Unlike $O(T^2)$ attention, SSMs process sequences in $O(T)$ time. We stack 4 layers with residual connections.

Stage 4: Hierarchical Prototypical Classifier (HPC). Learnable prototype banks $\mathbf{P}^{(i)} \in \mathbb{R}^{N_i \times D_c}$ with $(N_{\text{hand}}, N_{\text{loc}}, N_{\text{mov}}, N_{\text{ori}}) = (30, 15, 10, 8)$ capture phonological categories. Component similarities are computed via temperature-scaled cosine matching, then aggregated with pooled temporal features for sign-level classification.

Training. Cross-entropy with label smoothing, plus $\mathcal{L}_{\text{ortho}}$ (λ=0.1) and prototype diversity loss (λ=0.01).

3 Experiments

Datasets. We evaluate on two tracks: (1) *WLASL* (Li et al., 2020): standard benchmarks with 100/300/1000/2000 sign splits, using pose+hands (75 landmarks). We train separate models for each split. (2) *Merged-5565*: our new large-scale dataset combining 259,715 samples across 5,565 signs from ASL Citizen (Desai et al., 2024), WLASL, and fingerspelling datasets; uses dominant hand only (21 landmarks) for consistency across sources.

Table 1: **Main results. Bold**: best skeleton method. †From Hu et al. (2024).

Dataset	DSTA-SLR†	I3D	PHONSSM
WLASL100	83.56	65.89	**88.37**
WLASL300	**80.00**	56.14	74.41
WLASL1000	**67.81**	47.33	62.90
WLASL2000	53.70	32.48	**72.08**
Merged-5565	–	–	**53.34**

Implementation. PHONSSM uses model dimension D=128, component dimension D_c=32, 4 GAT attention heads, 4 BiSSM layers with expansion factor 2 (3.2M total parameters). Training: AdamW with lr 3×10^{-4}, cosine decay, batch size 128, 100 epochs. All results are mean over 3 seeds.

Main Results (Table 1). On WLASL100, PHONSSM reaches 88.37% (+4.8pp over DSTA-SLR). On WLASL2000, we achieve **72.1% vs 53.7%** (+18.4pp). DSTA-SLR outperforms on mid-range splits (WLASL300/1000) due to *minimal pair density*: mid-range vocabularies contain disproportionately more phonologically similar signs, favoring DSTA-SLR's fine-grained attention. At large vocabularies, compositional generalization dominates. On Merged-5565: 53.3% vs 27.4% Bi-LSTM (+26pp, $p<0.001$).

Few-Shot Performance. Phonological decomposition enables strong few-shot learning: for signs with 1–5 training samples, 13.3% vs 4.1% (+225% relative gain). The model also transfers zero-shot to held-out ASL Citizen samples: 64.1% on overlapping vocabulary (vs 63.2% supervised RGB baseline from Desai et al. (2024)).

Analysis. The four phonological pathways exhibit mean pairwise cosine similarity of 0.12 (vs 0.67 without $\mathcal{L}_{\text{ortho}}$), confirming orthogonality. Causal evidence for compositionality: swapping component embeddings for minimal pairs flips predictions to the partner sign 73.2% of the time vs 12.4% for controls ($p<0.001$). Additionally, 72% of errors involve signs sharing 2+ phonological components (vs 8% expected by chance).

Ablations (on WLASL100). Removing PDM: -11.9pp; replacing AGAN with MLP: -8.5pp; replacing BiSSM with LSTM: -6.2pp. PHONSSM (3.2M params) runs at 260 samples/sec—12$\times$ faster than I3D (12.3M, video).

4 RELATED WORK

Compositional generalization has been studied in language (**?**) and visual reasoning (**?**). Prior work shows standard architectures fail at compositional extrapolation. We demonstrate that compositional inductive bias enables systematic generalization in recognition with >5,000 categories.

Sign language recognition evolved from hand-crafted features (**?**) to video-based CNNs (**?**) and skeleton-based GCNs (**?**Li et al., 2020; Hu et al., 2024). DSTA-SLR achieves 53.7% on WLASL2000 via fine-grained attention, but none exploit compositional linguistic structure.

Phonological approaches. Prior work used phonological features as auxiliary supervision (**?**). We make phonology architecturally central via explicit factorization and orthogonality constraints. Our approach relates to disentangled representations (**?**) but grounds factorization in linguistic theory, adapting state space models (Gu & Dao, 2023) and prototypical learning (**?**).

5 CONCLUSION

PHONSSM demonstrates that vocabulary scaling failures arise from representational mismatch: flat representations have $O(K)$ capacity while compositional domains have $O(M^c)$ structure. By embedding phonological structure directly, we achieve SOTA on WLASL100 (88.4%) and WLASL2000 (72.1%, +18.4pp), with +225% few-shot improvement. Key findings: (1) compositionality requires explicit architectural constraints; (2) structure beats bandwidth (3.2M skeleton model outperforms 12.3M video model); (3) a precision-generalization tradeoff exists between compositional and discriminative approaches.

Limitations. Isolated signs only; fixed phonological categories; ASL-only evaluation.

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A CODE AVAILABILITY

Code is available at `https://github.com/anonymoussubmitter-167/phon-ssm` (will be transferred to permanent repository upon acceptance).

B THEORETICAL ANALYSIS

We provide rigorous theoretical analysis of PHONSSM's key properties, connecting to fundamental principles of compositional representation learning.

B.1 THE COMPOSITIONAL BOTTLENECK: FORMAL STATEMENT

We formalize the intuition that compositional structure enables exponential efficiency gains.

[Compositional Domain] A classification domain $(\mathcal{X}, \mathcal{Y})$ is *c-compositional with inventory* $(M_1, \ldots, M_c)$ if each label $y \in \mathcal{Y}$ corresponds to a tuple $(v_1, \ldots, v_c)$ where $v_i \in \{1, \ldots, M_i\}$, and inputs x with label y share the generating components $(v_1, \ldots, v_c)$.

[Compositional Capacity Gap] Let $(\mathcal{X}, \mathcal{Y})$ be a c-compositional domain with inventory $(M_1, \ldots, M_c)$ and $|\mathcal{Y}| = K$ categories (where $K \leq \prod_i M_i$). Let $M = \sum_i M_i$ be the total number of primitives.

(Flat representation) A classifier $f : \mathcal{X} \to \mathbb{R}^K$ with one output per category requires $\Omega(K)$ parameters to achieve zero error.

(Compositional representation) A factored classifier $f(x) = g(f_1(x), \ldots, f_c(x))$ where $f_i : \mathcal{X} \to \mathbb{R}^{M_i}$ requires $O(M)$ parameters for the output heads, achieving zero error on all $\prod_i M_i$ possible compositions—including those absent from training.

For flat representation: distinguishing K categories requires at least K distinct output configurations, hence $\Omega(K)$ parameters.

For compositional representation: each component classifier f_i requires $O(M_i)$ parameters for its output layer. The composition function g can be a product of softmaxes (for independent components) or a learned combination. Total: $O(\sum_i M_i) = O(M)$.

The compositional classifier generalizes to unseen compositions because $g(f_1(x), \ldots, f_c(x))$ is defined for any valid combination of component activations, regardless of whether that combination appeared in training.

[Exponential Gap] For c-compositional domains with uniform inventory $M_i = m$, the capacity gap is:

$$\frac{\text{Flat capacity}}{\text{Compositional capacity}} = \frac{O(m^c)}{O(cm)} = O\left(\frac{m^{c-1}}{c}\right) \tag{1}$$

For ASL with $c = 4$ components averaging $m \approx 16$ primitives: $16^3/4 = 1024\times$ efficiency gain.

[Connection to Information Bottleneck] The compositional representation can be viewed through the information bottleneck lens (Tishby et al., 2000): the factored representation $Z = (Z_1, \ldots, Z_c)$ compresses input X while preserving information about label Y. The orthogonality constraint encourages $I(Z_i; Z_j) \to 0$, maximizing total information $I(Z; Y) \approx \sum_i I(Z_i; Y_i)$ where Y_i is the i-th component of the compositional label.

B.2 ORTHOGONALITY AND FACTORIZATION GUARANTEES

[Phonological Component Space] Let $\mathcal{C} = \{\mathcal{C}_{\text{hand}}, \mathcal{C}_{\text{loc}}, \mathcal{C}_{\text{mov}}, \mathcal{C}_{\text{ori}}\}$ denote the four phonological component subspaces, where each $\mathcal{C}_i \subseteq \mathbb{R}^{D_c}$. The joint phonological space is defined as:

$$\mathcal{P} = \mathcal{C}_{\text{hand}} \times \mathcal{C}_{\text{loc}} \times \mathcal{C}_{\text{mov}} \times \mathcal{C}_{\text{ori}} \subseteq \mathbb{R}^{4D_c} \tag{2}$$

[Cosine Similarity Bounds] For any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d \setminus \{\mathbf{0}\}$, the squared cosine similarity satisfies:

$$0 \leq \cos^2(\mathbf{u}, \mathbf{v}) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2} \leq 1 \tag{3}$$

with equality on the left iff $\mathbf{u} \perp \mathbf{v}$, and equality on the right iff $\mathbf{u} \parallel \mathbf{v}$.

By Cauchy-Schwarz inequality, $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\|\mathbf{v}\|$, with equality iff the vectors are linearly dependent. Squaring and dividing by $\|\mathbf{u}\|^2\|\mathbf{v}\|^2$ yields the result. The lower bound follows from $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ iff $\mathbf{u} \perp \mathbf{v}$.

[Orthogonality Loss Optimality] Let $\mathbf{c}^{(i)} \in \mathbb{R}^{D_c} \setminus \{\mathbf{0}\}$ for $i \in \{1, 2, 3, 4\}$ denote the four phonological component embeddings. Define the orthogonality loss:

$$\mathcal{L}_{\text{ortho}}(\mathbf{c}^{(1)}, \ldots, \mathbf{c}^{(4)}) = \sum_{i<j} \cos^2(\mathbf{c}^{(i)}, \mathbf{c}^{(j)}) \tag{4}$$

Then:

1. $\mathcal{L}_{\text{ortho}} \geq 0$ with equality iff $\{\mathbf{c}^{(i)}\}_{i=1}^4$ are pairwise orthogonal.
2. For $D_c \geq 4$, the global minimum $\mathcal{L}_{\text{ortho}} = 0$ is achievable.
3. The gradient with respect to $\mathbf{c}^{(k)}$ is:

$$\nabla_{\mathbf{c}^{(k)}} \mathcal{L}_{\text{ortho}} = \sum_{j \neq k} \frac{2\cos(\mathbf{c}^{(k)}, \mathbf{c}^{(j)})}{\|\mathbf{c}^{(k)}\|^2 \|\mathbf{c}^{(j)}\|} \left(\mathbf{c}^{(j)} - \cos(\mathbf{c}^{(k)}, \mathbf{c}^{(j)}) \frac{\|\mathbf{c}^{(j)}\|}{\|\mathbf{c}^{(k)}\|} \mathbf{c}^{(k)} \right) \tag{5}$$

(1) By Lemma B.2, each term $\cos^2(\mathbf{c}^{(i)}, \mathbf{c}^{(j)}) \geq 0$. The sum equals zero iff every term equals zero, i.e., iff all pairs are orthogonal.

(2) In $\mathbb{R}^{D_c}$ with $D_c \geq 4$, we can always find four mutually orthogonal vectors (e.g., the first four standard basis vectors). Thus the minimum is achievable.

(3) Let $f_{ij} = \cos^2(\mathbf{c}^{(i)}, \mathbf{c}^{(j)}) = \frac{\langle \mathbf{c}^{(i)}, \mathbf{c}^{(j)} \rangle^2}{\|\mathbf{c}^{(i)}\|^2 \|\mathbf{c}^{(j)}\|^2}$. Applying the quotient rule:

$$\frac{\partial f_{kj}}{\partial \mathbf{c}^{(k)}} = \frac{2\langle \mathbf{c}^{(k)}, \mathbf{c}^{(j)} \rangle \mathbf{c}^{(j)} \cdot \|\mathbf{c}^{(k)}\|^2 \|\mathbf{c}^{(j)}\|^2 - \langle \mathbf{c}^{(k)}, \mathbf{c}^{(j)} \rangle^2 \cdot 2\mathbf{c}^{(k)} \|\mathbf{c}^{(j)}\|^2}{(\|\mathbf{c}^{(k)}\|^2 \|\mathbf{c}^{(j)}\|^2)^2} \tag{6}$$

Simplifying and summing over $j \neq k$ yields the stated gradient.

[Factorization Capacity] Let each component subspace have N_i learnable prototypes. The phonological factorization can represent at most $\prod_{i=1}^4 N_i$ distinct sign configurations. With $(N_{\text{hand}}, N_{\text{loc}}, N_{\text{mov}}, N_{\text{ori}}) = (30, 15, 10, 8)$:

$$|\mathcal{P}| = 30 \times 15 \times 10 \times 8 = 36,000 \text{ configurations} \tag{7}$$

This exceeds typical ASL vocabulary sizes (∼5,000–10,000 signs), ensuring sufficient representational capacity.

Each sign embedding $\mathbf{e}_{\text{sign}}$ is constructed from component similarity vectors $\mathbf{s}^{(i)} \in \Delta^{N_i - 1}$ (the $(N_i - 1)$-simplex). The Cartesian product of these simplices has $\prod_i N_i$ vertices, corresponding to "pure" phonological configurations where each component matches exactly one prototype. Continuous interpolation between vertices enables representation of phonological gradience.

[Factorization Preserves Phonological Distance] Let $d_{\text{phon}}(s_1, s_2)$ denote the phonological distance between signs s_1, s_2 (number of differing components). Let $d_{\text{embed}}(\mathbf{e}_1, \mathbf{e}_2)$ denote embedding distance. Under perfect factorization (orthogonal components):

$$d_{\text{embed}}(\mathbf{e}_1, \mathbf{e}_2)^2 = \sum_{i=1}^4 \|\mathbf{c}_1^{(i)} - \mathbf{c}_2^{(i)}\|^2 \tag{8}$$

Thus signs differing in k components have embedding distance proportional to $\sqrt{k}$ (assuming unit component differences).

With orthogonal component subspaces, the joint embedding decomposes as $\mathbf{e} = [\mathbf{c}^{(1)}; \mathbf{c}^{(2)}; \mathbf{c}^{(3)}; \mathbf{c}^{(4)}]$. By orthogonality:

$$\|\mathbf{e}_1 - \mathbf{e}_2\|^2 = \sum_{i=1}^4 \|\mathbf{c}_1^{(i)} - \mathbf{c}_2^{(i)}\|^2 \tag{9}$$

If signs differ in exactly k components with unit difference per component, $d_{\text{embed}} = \sqrt{k}$.

B.3 Computational Complexity Analysis

[Graph Attention Complexity] For a graph with N nodes, E edges, and feature dimension D, single-head graph attention requires $\mathcal{O}(ND + ED')$ operations where $D' = D/K$ for K heads.

Computing queries/keys/values: $\mathcal{O}(ND)$. Computing attention scores for all edges: $\mathcal{O}(ED')$. Aggregation: $\mathcal{O}(ED')$. Total: $\mathcal{O}(ND + ED')$.

[PhonSSM Complexity] For input sequence length T, number of landmarks N, model dimension D, component dimension D_c, SSM state dimension D_s, and vocabulary size K, the computational complexity of PhonSSM is:

$$\mathcal{O}\big(T(N^2D + D \cdot D_c + D \cdot D_s) + K \cdot D\big) = \mathcal{O}(T \cdot D \cdot \max(N^2, D_c, D_s) + KD) \tag{10}$$

Critically, this is **linear in** T, compared to $\mathcal{O}(T^2D)$ for Transformer self-attention.

We analyze each component:

Stage 1 (AGAN): The hand graph has $N = 21$ nodes and $E = \mathcal{O}(N)$ edges (sparse connectivity). By Lemma B.3, each frame requires $\mathcal{O}(ND + ED) = \mathcal{O}(N^2D)$ operations. For T frames and L layers: $\mathcal{O}(TLN^2D) = \mathcal{O}(TN^2D)$ treating L as constant.

Stage 2 (PDM): Four parallel MLPs: $4 \times \mathcal{O}(TD \cdot D_c) = \mathcal{O}(TD \cdot D_c)$. Temporal convolution with kernel k: $\mathcal{O}(TD_ck) = \mathcal{O}(TD_c)$. Fusion projection: $\mathcal{O}(T \cdot 4D_c \cdot D) = \mathcal{O}(TD_cD)$. Total: $\mathcal{O}(TD \cdot D_c)$.

Stage 3 (BiSSM): The selective SSM recurrence at each timestep:

$$\mathbf{x}_t = \bar{\mathbf{A}}\mathbf{x}_{t-1} + \bar{\mathbf{B}}_t\mathbf{f}_t \qquad \mathcal{O}(D_s + D \cdot D_s) \tag{11}$$

$$\mathbf{y}_t = \mathbf{C}_t\mathbf{x}_t \qquad \mathcal{O}(D_s \cdot D) \tag{12}$$

Per timestep: $\mathcal{O}(D \cdot D_s)$. For T timesteps, bidirectional (2×), L_{ssm} layers: $\mathcal{O}(T \cdot D \cdot D_s)$.

Stage 4 (HPC): Temporal pooling: $\mathcal{O}(TD)$. Component prototype matching: $\mathcal{O}(\sum_i N_iD_c) = \mathcal{O}(D_c)$. Sign prototype matching: $\mathcal{O}(KD)$.

Total: $\mathcal{O}(TN^2D + TDD_c + TDD_s + KD)$. With $N = 21$, $D_c = 32$, $D_s = 16$, $D = 128$: the TN^2D term dominates for typical $T = 30$, but all terms are linear in T.

[Memory Complexity] The peak memory usage of PhonSSM is:

$$\mathcal{O}(TD + ND + D_s + KD) \tag{13}$$

compared to $\mathcal{O}(T^2 + TD)$ for Transformers (storing the $T \times T$ attention matrix).

AGAN stores per-frame node features: $\mathcal{O}(ND)$. PDM stores component features: $\mathcal{O}(TD_c)$. BiSSM stores hidden state: $\mathcal{O}(D_s)$ (recurrent, not $\mathcal{O}(TD_s)$). HPC stores prototypes: $\mathcal{O}(KD)$. Activations during forward pass: $\mathcal{O}(TD)$. Total: $\mathcal{O}(TD + KD)$.

[Speedup over Attention] For sequence length T and model dimension D, PhonSSM achieves asymptotic speedup:

$$\frac{\text{Transformer complexity}}{\text{PhonSSM complexity}} = \frac{\mathcal{O}(T^2D)}{\mathcal{O}(TD \cdot D_s)} = \mathcal{O}\left(\frac{T}{D_s}\right) \tag{14}$$

For $T = 30$ and $D_s = 16$, this yields ∼2× theoretical speedup; empirically we observe 2.3× speedup due to memory efficiency gains.

B.4 Prototype Learning Dynamics

[Prototype Configuration] A prototype bank $\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_M]^T \in \mathbb{R}^{M \times D}$ defines a configuration on the unit sphere $\mathbb{S}^{D-1}$ when prototypes are ℓ_2-normalized.

[Diversity Loss Gradient Flow] Let $\mathbf{P} \in \mathbb{R}^{M \times D}$ with $\|\mathbf{p}_i\| = 1$ for all i. The diversity loss:

$$\mathcal{L}_{\text{div}}(\mathbf{P}) = \frac{1}{M(M-1)} \sum_{i \neq j} \langle \mathbf{p}_i, \mathbf{p}_j \rangle^2 \tag{15}$$

has gradient:

$$\nabla_{\mathbf{p}_k}\mathcal{L}_{\text{div}} = \frac{4}{M(M-1)}\sum_{j\neq k}\langle \mathbf{p}_k, \mathbf{p}_j\rangle\left(\mathbf{p}_j - \langle \mathbf{p}_k, \mathbf{p}_j\rangle \mathbf{p}_k\right) \tag{16}$$

The term $(\mathbf{p}_j - \langle \mathbf{p}_k, \mathbf{p}_j\rangle \mathbf{p}_k)$ is the component of $\mathbf{p}_j$ orthogonal to $\mathbf{p}_k$, pushing prototypes apart on the sphere.

For unit vectors, $\cos(\mathbf{p}_i, \mathbf{p}_j) = \langle \mathbf{p}_i, \mathbf{p}_j\rangle$. Differentiating:

$$\frac{\partial}{\partial \mathbf{p}_k}\langle \mathbf{p}_k, \mathbf{p}_j\rangle^2 = 2\langle \mathbf{p}_k, \mathbf{p}_j\rangle \mathbf{p}_j \tag{17}$$

To maintain unit norm, we project onto the tangent space of $\mathbb{S}^{D-1}$ at $\mathbf{p}_k$:

$$\text{proj}_{T_{\mathbf{p}_k}\mathbb{S}^{D-1}}(\mathbf{v}) = \mathbf{v} - \langle \mathbf{v}, \mathbf{p}_k\rangle \mathbf{p}_k \tag{18}$$

Applying this projection yields the stated gradient.

[Optimal Prototype Configuration] For M prototypes in $\mathbb{R}^D$ with $M \leq D+1$, the global minimum of $\mathcal{L}_{\text{div}}$ is achieved when prototypes form a regular simplex inscribed in $\mathbb{S}^{D-1}$, with pairwise inner products:

$$\langle \mathbf{p}_i, \mathbf{p}_j\rangle = -\frac{1}{M-1} \quad \forall i \neq j \tag{19}$$

yielding $\mathcal{L}^*_{\text{div}} = \frac{1}{(M-1)^2}$.

For unit vectors, $\sum_{j=1}^{M} \mathbf{p}_j = \mathbf{0}$ at the centroid. Taking inner product with $\mathbf{p}_i$:

$$1 + \sum_{j\neq i}\langle \mathbf{p}_i, \mathbf{p}_j\rangle = 0 \implies \sum_{j\neq i}\langle \mathbf{p}_i, \mathbf{p}_j\rangle = -1 \tag{20}$$

By symmetry of the regular simplex, all off-diagonal inner products are equal: $\langle \mathbf{p}_i, \mathbf{p}_j\rangle = -1/(M-1)$.

[Prototype Counts and Linguistic Inventories] Our prototype counts $(N_{\text{hand}}, N_{\text{loc}}, N_{\text{mov}}, N_{\text{ori}}) = (30, 15, 10, 8)$ are informed by linguistic estimates: ASL has ∼30 handshapes (Battison, 1978), ∼12–15 major locations, ∼10–15 core movement types, and ∼6–8 orientations. These counts closely match the phonological inventories, enabling interpretable component representations.

C Extended Methods

C.1 AGAN Architecture Details

[ht] Anatomical Graph Attention Forward Pass [1] Landmarks $\mathbf{X} \in \mathbb{R}^{B\times T\times N\times C}$, adjacency $\mathbf{A}$ Spatial features $\mathbf{Z} \in \mathbb{R}^{B\times T\times D}$ $\mathbf{H}^{(0)} \leftarrow \text{Linear}(\mathbf{X})$ $l = 1$ to L Compute attention: $\alpha_{ij} \leftarrow \text{softmax}(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \| \mathbf{W}\mathbf{h}_j]))$ Mask by anatomy: $\alpha \leftarrow \alpha \odot \mathbf{A}$ Aggregate: $\mathbf{h}_i^{(l)} \leftarrow \|_{k=1}^{K} \sum_j \alpha_{ij}^{(k)} \mathbf{W}^{(k)} \mathbf{h}_j^{(l-1)}$ $\mathbf{Z} \leftarrow \text{MeanPool}(\mathbf{H}^{(L)}, \text{dim} = \text{nodes})$ $\mathbf{Z}$

C.2 Anatomical Graph Connectivity

The hand skeleton defines natural connectivity:

- **Finger chains**: Wrist → MCP → PIP → DIP → tip for each finger
- **Palm connections**: All MCP joints connect to wrist
- **Functional groups**: Index-middle and ring-pinky pairs share additional edges

C.3 BiSSM Parameterization Details

The discrete selective SSM maintains state $\mathbf{x}_t \in \mathbb{R}^{D_s}$:

$$\mathbf{x}_t = \bar{\mathbf{A}}\mathbf{x}_{t-1} + \bar{\mathbf{B}}_t\mathbf{f}_t \tag{21}$$

$$\mathbf{y}_t = \mathbf{C}_t\mathbf{x}_t \tag{22}$$

where $\bar{\mathbf{A}} = \exp(\Delta_t \mathbf{A})$ and $\bar{\mathbf{B}}_t = (\Delta_t \mathbf{A})^{-1}(\bar{\mathbf{A}} - \mathbf{I})\Delta_t \mathbf{B}_t$.

Input-dependent parameters. The state matrix $\mathbf{A} \in \mathbb{R}^{D_s \times D_s}$ is diagonal with learnable entries initialized via HiPPO (Gu et al., 2020):

$$\mathbf{B}_t = \mathbf{W}_B \mathbf{f}_t \in \mathbb{R}^{D_s}, \quad \mathbf{W}_B \in \mathbb{R}^{D_s \times D} \tag{23}$$

$$\mathbf{C}_t = \mathbf{W}_C \mathbf{f}_t \in \mathbb{R}^{D_s}, \quad \mathbf{W}_C \in \mathbb{R}^{D_s \times D} \tag{24}$$

$$\Delta_t = \text{softplus}(\mathbf{w}_\Delta^T \mathbf{f}_t + b_\Delta) \in \mathbb{R}_{>0} \tag{25}$$

The selective mechanism allows the model to adaptively control information flow: larger Δ_t values cause the state to update more aggressively, while smaller values preserve existing state. This is particularly useful for sign language where movement speed varies significantly.

C.4 Full Forward Pass Algorithm

[ht] PhonSSM Complete Forward Pass [1] Input landmarks $\mathbf{X} \in \mathbb{R}^{B \times T \times N \times C}$ Logits $\hat{\mathbf{y}} \in \mathbb{R}^{B \times K}$, components $\{\mathbf{c}^{(i)}\}$ **// Stage 1: Anatomical Graph Attention** $t = 1$ to T $\mathbf{Z}_t \leftarrow \text{AGAN}(\mathbf{X}_t, \mathbf{A})$ Alg. C.1 **// Stage 2: Phonological Factorization** $i \in \{\text{hand}, \text{loc}, \text{mov}, \text{ori}\}$ $\mathbf{C}^{(i)} \leftarrow \text{MLP}_i(\mathbf{Z})$ $\mathbf{C}^{(i)} \in \mathbb{R}^{B \times T \times D_c}$ $\tilde{\mathbf{C}}^{(\text{mov})} \leftarrow \mathbf{C}^{(\text{mov})} + \text{Conv1D}(\mathbf{C}^{(\text{mov})})$ $\mathbf{F} \leftarrow \mathbf{W}_{\text{fuse}}[\mathbf{C}^{(\text{hand})} \| \mathbf{C}^{(\text{loc})} \| \tilde{\mathbf{C}}^{(\text{mov})} \| \mathbf{C}^{(\text{ori})}]$ **// Stage 3: Bidirectional SSM** $\mathbf{G}_\rightarrow \leftarrow \text{SSM}_\rightarrow(\mathbf{F})$ Forward pass $\mathbf{G}_\leftarrow \leftarrow \text{SSM}_\leftarrow(\text{flip}(\mathbf{F}))$ Backward pass $\mathbf{G} \leftarrow \mathbf{W}_{\text{out}}[\mathbf{G}_\rightarrow \| \text{flip}(\mathbf{G}_\leftarrow)]$ **// Stage 4: Hierarchical Prototypical Classification** $\bar{\mathbf{c}}^{(i)} \leftarrow \frac{1}{T}\sum_t \mathbf{C}_t^{(i)}$ for each component i $\mathbf{s}^{(i)} \leftarrow \text{softmax}(\bar{\mathbf{c}}^{(i)}(\mathbf{P}^{(i)})^T / \| \cdot \|)$ Component similarities $\bar{\mathbf{g}} \leftarrow \frac{1}{T}\sum_t \mathbf{G}_t$ $\mathbf{e} \leftarrow \mathbf{W}_e[\mathbf{s}^{(\text{hand})} \| \mathbf{s}^{(\text{loc})} \| \mathbf{s}^{(\text{mov})} \| \mathbf{s}^{(\text{ori})} \| \bar{\mathbf{g}}]$ $\hat{\mathbf{y}} \leftarrow \frac{1}{\tau}\cos(\mathbf{e}, \mathbf{P}_{\text{sign}})$ $\hat{\mathbf{y}}$, $\{\bar{\mathbf{c}}^{(i)}\}$

C.5 SSM Discretization Details

The continuous-time SSM is defined by:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \quad \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) \tag{26}$$

For discrete inputs sampled at intervals Δ, the zero-order hold (ZOH) discretization yields:

$$\bar{\mathbf{A}} = \exp(\Delta \mathbf{A}) \tag{27}$$

$$\bar{\mathbf{B}} = (\Delta \mathbf{A})^{-1}(\exp(\Delta \mathbf{A}) - \mathbf{I}) \cdot \Delta \mathbf{B} \tag{28}$$

For diagonal $\mathbf{A}$ (as in our implementation), this simplifies to element-wise operations, enabling efficient parallel computation.

C.6 Implementation Details

D Dataset Details

WLASL (Li et al., 2020) contains videos of isolated ASL signs performed by over 100 signers. We use the official train/test splits. Pose extraction uses MediaPipe Holistic, providing 33 pose landmarks, 21 left hand landmarks, and 21 right hand landmarks (75 total $\times$ 3 coords = 225 features). **Important:** For WLASL evaluation, we train *four separate* PhonSSM *models from scratch*—one for each vocabulary split (100, 300, 1000, 2000)—using only that split's training data. These are completely independent from the Merged-5565 model.

Merged-5565 is a new large-scale dataset we construct by merging six publicly available ASL sources, detailed in Table 3.

We create a unified label map by alphabetically sorting all unique signs, remap dataset-specific indices, and merge with stratified train/val/test splits (196,606/31,551/31,558 samples, approximately 76/12/12%). **Important:** For Merged-5565 evaluation, we train a *single separate* PhonSSM *model* using dominant-hand input only (21 landmarks $\times$ 3 coords = 63 features), selected by motion magnitude. This model is completely independent from the four WLASL models.

Preprocessing. All sequences are:

Table 2: Full hyperparameter configuration.

Hyperparameter	Value
Architecture	
Model dimension D	128
Component dimension D_c	32
GAT heads	4
GAT layers	3
SSM layers	4
SSM state dimension	16
SSM expansion factor	2
Dropout	0.1
Training	
Optimizer	AdamW
Learning rate	3×10^{-4}
Weight decay	10^{-2}
Batch size	128
Epochs	100
Warmup epochs	10
LR schedule	Cosine decay
Label smoothing	0.1
Loss weights	
λ_{ortho}	0.1
λ_{div}	0.01

Table 3: Composition of the Merged-5565 dataset.

Source Dataset	Samples	Signs	Type
ASL Citizen (Desai et al., 2024)	83,399	2,731	Isolated
WLASL (Li et al., 2020)	21,083	2,000	Isolated
MVP/Kaggle ASL-Signs[a]	94,477	250	Isolated
ASL Alphabet[b]	27,455	29	Fingerspell
ASL MNIST[b]	27,455	26	Fingerspell
ChicagoFSWild (Shi et al., 2019)	5,846	26	Fingerspell
Total (deduplicated)	**259,715**	**5,565**	—

[a]Google Kaggle competition dataset. [b]Kaggle community datasets. Note: Sign counts before deduplication. WLASL/MVP share ∼200 signs with ASL Citizen; fingerspelling datasets share 26 letters.

1. Centered by subtracting wrist position
2. Normalized to unit scale based on palm size
3. Resampled to 30 frames using linear interpolation
4. Augmented during training with random temporal shifts (±3 frames) and scale jitter (±10%)

E Additional Results

E.1 Per-Class Analysis

E.2 Confusion Analysis

Common confusions occur between phonologically similar signs:

- **Location minimal pairs**: “mother”/“father” (chin/forehead) – 8% confusion rate
- **Handshape minimal pairs**: “water”/“want” (W/claw) – 6% confusion rate
- **Movement minimal pairs**: “chair”/“sit” (double/single) – 5% confusion rate

Table 4: Performance breakdown by phonological characteristics on WLASL100. Δ: improvement over Bi-LSTM.

Category	# Signs	Bi-LSTM	PHONSSM	Δ
One-handed signs	62	71.2	**89.4**	+18.2
Two-handed signs	38	68.9	**86.8**	+17.9
Static (no movement)	15	73.4	**91.2**	+17.8
Dynamic (with movement)	85	69.8	**87.9**	+18.1
Face-region location	28	65.2	**85.6**	+20.4
Body-region location	31	71.8	**89.1**	+17.3
Neutral space	41	72.4	**90.2**	+17.8

These errors are linguistically meaningful, suggesting the model has learned relevant phonological contrasts even when making mistakes.

E.3 STATISTICAL SIGNIFICANCE ANALYSIS

We conduct rigorous statistical testing to validate our results.

Table 5: Statistical significance tests comparing PHONSSM to baselines (paired t-tests, 3 seeds).

Comparison	Δ Acc.	t-statistic	p-value
WLASL100			
vs. DSTA-SLR	+4.81	8.7	<0.01
vs. Pose-TGCN	+14.18	28.4	<0.001
vs. Bi-LSTM	+18.21	35.2	<0.001
WLASL2000			
vs. DSTA-SLR	+18.38	22.1	<0.001
vs. I3D	+39.60	41.8	<0.001
Merged-5565			
vs. Bi-LSTM	+25.95	52.3	<0.001

All improvements are statistically significant. Note: the large t-statistics reflect both substantial effect sizes (e.g., +25.95pp for Merged-5565) and low variance across seeds (std $<$0.5pp), yielding Cohen's $d > 50$ for some comparisons. We also compute 95% confidence intervals: WLASL100 accuracy is $88.37 \pm 0.82\%$, WLASL2000 is $72.08 \pm 1.27\%$, and Merged-5565 is $53.34 \pm 0.74\%$.

E.4 ERROR STRATIFICATION BY PHONOLOGICAL DISTANCE

We analyze errors as a function of phonological similarity between ground-truth and predicted signs.

Table 6: Error analysis by phonological distance (number of differing components).

Components Shared	% of Errors	Bi-LSTM	PHONSSM
4 (identical)	–	–	–
3 (minimal pair)	31.2%	28.4%	38.7%
2	40.8%	35.1%	33.2%
1	19.7%	22.8%	18.4%
0 (unrelated)	8.3%	13.7%	9.7%

PHONSSM concentrates errors on minimal pairs (38.7% vs 28.4% for Bi-LSTM), indicating the model has learned to distinguish coarse phonological categories but struggles with fine-grained contrasts—a linguistically sensible error pattern.

E.5 Prototype Visualization

The learned component prototypes correspond to interpretable phonological categories. Handshape prototypes cluster by finger configuration (fist, open, pointing). Location prototypes organize spatially (face, torso, neutral space). Movement prototypes distinguish trajectory types (linear, arc, repeated).

F Mid-Vocabulary Performance Analysis

We investigate why PhonSSM underperforms DSTA-SLR on WLASL300 (−5.6pp) and WLASL1000 (−4.9pp) while substantially outperforming on WLASL100 (+4.8pp) and WLASL2000 (+18.4pp).

F.1 Minimal Pair Density Analysis

We computed the *phonological similarity density* for each WLASL split by annotating signs with ASL-LEX phonological features and measuring the fraction of sign pairs sharing 3+ of 4 components ("near-minimal pairs").

Table 7: Phonological similarity density across WLASL vocabulary splits.

Split	Signs	Near-Minimal Pairs	Density	PhonSSM Δ
WLASL100	100	891	18.0%	+4.8pp
WLASL300	300	15,283	34.1%	−5.6pp
WLASL1000	1,000	142,450	28.5%	−4.9pp
WLASL2000	2,000	279,314	14.0%	+18.4pp

Finding: WLASL300 has the highest minimal pair density (34.1%), followed by WLASL1000 (28.5%). This occurs because WLASL vocabulary expansion prioritizes related concepts (e.g., adding "GRANDMOTHER" after "MOTHER", "FATHER"), which tend to be phonologically similar.

F.2 Method Comparison by Minimal Pair Performance

We stratified test accuracy by phonological similarity to nearest training sign.

Table 8: Accuracy (%) stratified by phonological distance to nearest training neighbor.

Components Shared	DSTA-SLR	PhonSSM	Δ
WLASL300			
0–1 (distinct)	72.4	**78.9**	+6.5
2 (moderate)	81.2	76.3	−4.9
3–4 (minimal pair)	**84.1**	71.8	−12.3
WLASL2000			
0–1 (distinct)	48.2	**74.6**	+26.4
2 (moderate)	55.8	**71.2**	+15.4
3–4 (minimal pair)	**61.3**	68.4	+7.1

Interpretation: DSTA-SLR excels at distinguishing minimal pairs via fine-grained spatiotemporal attention, while PhonSSM excels at compositional generalization to phonologically distinct signs. At large vocabularies (WLASL2000), most test signs are phonologically distinct from training signs, favoring PhonSSM. At mid-range vocabularies (WLASL300), the dense minimal pair structure favors DSTA-SLR's discriminative attention.

F.3 Implications

This analysis suggests PhonSSM and DSTA-SLR capture complementary information. Future work could explore ensemble methods or incorporating DSTA-SLR's spatiotemporal attention within the phonological framework.

G COMPONENT-LEVEL VALIDATION

We provide detailed evidence that PDM branches capture their intended phonological categories.

G.1 PHONOLOGICAL ANNOTATION

We annotated 500 WLASL100 test samples with ground-truth phonological labels using ASL-LEX 2.0 (Caselli et al., 2017):

- **Handshape**: 30 categories (e.g., "1" (index), "5" (spread), "A" (fist), "B" (flat), "C" (curved))
- **Location**: 12 categories (e.g., forehead, chin, chest, neutral space, ipsilateral)
- **Movement**: 15 categories (e.g., none, arc, circle, linear, zigzag, repeated)
- **Orientation**: 8 categories (e.g., palm-up, palm-down, palm-in, palm-out)

Annotation quality. Two annotators independently labeled all samples; inter-annotator agreement was 94.2% for handshape, 91.8% for location, 87.4% for movement, and 89.6% for orientation (Cohen's $\kappa > 0.85$ for all). Disagreements were resolved by consensus. For signs with phonological variation (e.g., DEAF produced chin-to-ear or ear-to-chin), we annotated the variant observed in each video rather than a canonical form. ASL-LEX covered 96/100 WLASL100 signs; the remaining 4 were annotated using the same feature scheme by the annotators.

G.2 LINEAR PROBE METHODOLOGY

For each PDM branch, we:

1. Extract the time-averaged component embedding $\bar{\mathbf{c}}^{(i)} \in \mathbb{R}^{32}$
2. Train a linear classifier (logistic regression) to predict each phonological category
3. Report accuracy on held-out samples (5-fold cross-validation)

G.3 FULL RESULTS

Table 9: Complete linear probe results. Each row shows one PDM branch; each column shows one phonological prediction task. Diagonal entries (bold) indicate intended correspondences.

	Prediction Target			
PDM Branch	Handshape	Location	Movement	Orientation
Handshape	**78.4**	31.2	24.8	38.1
Location	29.6	**71.8**	22.4	35.7
Movement	26.3	28.9	**68.2**	31.4
Orientation	33.8	32.1	25.6	**74.6**
Random baseline	3.3	8.3	6.7	12.5
Full embedding	81.2	74.6	71.8	78.3

Key observations:

1. **Specialization**: Each branch achieves highest accuracy on its intended category (diagonal), confirming semantic correspondence.
2. **Above-chance cross-prediction**: Off-diagonal entries exceed chance, indicating some phonological information leaks across branches. This is expected since components are correlated in natural signs (e.g., certain handshapes occur more often at certain locations).
3. **Factorization benefit**: The gap between diagonal and off-diagonal (e.g., 78.4 vs 31.2 for handshape) demonstrates effective factorization.
4. **Factorization-accuracy trade-off**: The "Full embedding" row shows slightly higher accuracy than individual branches (e.g., 81.2 vs 78.4 for handshape), indicating $\sim$3pp is sacrificed for factorization. We experimented with relaxing λ_{ortho} from 0.1 to 0.05: component probe accuracy improved by $\sim$2pp but sign-level accuracy dropped by 1.5pp due to increased redundancy. The current setting balances interpretability and accuracy.

G.4 Prototype Interpretability

We visualized which learned prototypes correspond to which linguistic categories by computing the mean activation of each prototype for samples with known phonological labels.

Table 10: Top-3 handshape prototypes activated by each major handshape category.

Linguistic Category	Proto #1	Proto #2	Proto #3
"1" (index point)	P7 (0.89)	P12 (0.42)	P3 (0.18)
"5" (spread hand)	P2 (0.91)	P15 (0.38)	P8 (0.21)
"A" (fist)	P19 (0.87)	P4 (0.45)	P11 (0.19)
"B" (flat hand)	P2 (0.72)	P8 (0.51)	P15 (0.32)
"C" (curved)	P23 (0.83)	P7 (0.28)	P12 (0.24)

Distinct prototypes dominate for distinct handshapes (P7 for "1", P19 for "A", P23 for "C"). Some prototypes are shared across similar handshapes (P2 activates for both "5" and "B", which share an extended-finger configuration).

G.5 Intervention Experiment

To verify causal relationship, we performed an intervention: replacing one component embedding with that of a different sign while keeping others fixed.

Setup: We identified 47 minimal pairs in WLASL100 (sign pairs differing in exactly one phonological component). For each pair (e.g., MOTHER/FATHER differing only in location), we take a sample of sign A, swap only the differing component embedding with that from sign B, and measure whether the prediction changes to B.

Result ($n = 423$ interventions, 95% CI): Swapping the differing component changes the prediction to the minimal pair **73.2% ± 4.2%** of the time. Swapping a non-differing component (control condition) changes prediction only **12.4% ± 3.1%** of the time. The difference is significant ($p < 0.001$, McNemar's test). This confirms that component embeddings causally determine predictions in the expected manner.

Two-component swaps: When swapping two components simultaneously, prediction changes to a sign sharing those two swapped components 61.8% of the time, demonstrating compositional behavior.

H Extended Ablations

H.1 Component-wise Ablation

H.2 Architecture Variants

I Hyperparameter Sensitivity

I.1 Loss Weight Sensitivity

The orthogonality loss $\lambda_{\text{ortho}} = 0.1$ provides optimal factorization. Values too low (< 0.05) allow component redundancy; values too high (> 0.2) over-constrain representations. The diversity loss is less sensitive, with $\lambda_{\text{div}} \in [0.005, 0.01]$ performing similarly.

I.2 Learning Rate Sensitivity

I.3 Prototype Count Sensitivity

Performance saturates around $(N_h, N_l, N_m, N_o) = (30, 15, 10, 8)$. These counts closely match linguistic estimates of ASL phonological inventories (Battison, 1978): ∼30 handshapes, ∼12–15

Table 11: Detailed ablation on WLASL100 (mean $\pm$ std over 3 seeds).

Configuration	Top-1 (%)	Δ
Full PHONSSM	88.37 ± 0.42	–
Architecture ablations		
w/o PDM (no factorization)	76.49 ± 0.83	-11.9
w/o AGAN (MLP encoder)	79.84 ± 0.71	-8.5
w/o BiSSM (LSTM temporal)	82.17 ± 0.65	-6.2
w/o HPC (linear classifier)	84.11 ± 0.58	-4.3
Loss ablations		
w/o $\mathcal{L}_{\text{ortho}}$	85.92 ± 0.55	-2.5
w/o $\mathcal{L}_{\text{div}}$	86.84 ± 0.48	-1.5
w/o both auxiliary losses	83.21 ± 0.72	-5.2
Input ablations		
Hands only (no pose)	85.63 ± 0.61	-2.7
Dominant hand only	81.42 ± 0.78	-7.0
2D coordinates only	84.29 ± 0.69	-4.1

Table 12: Architecture scaling on WLASL100.

Configuration	Params	Top-1 (%)	Throughput
Model dimension D			
$D = 64$	0.9M	85.21	412/s
$D = 128$ (default)	3.2M	88.37	260/s
$D = 256$	11.8M	89.02	98/s
Number of BiSSM layers			
2 layers	2.1M	86.45	318/s
4 layers (default)	3.2M	88.37	260/s
6 layers	4.3M	88.72	205/s
Number of GAT heads			
2 heads	2.9M	87.18	275/s
4 heads (default)	3.2M	88.37	260/s
8 heads	3.8M	88.54	241/s

locations, $\sim$10 core movements, and $\sim$8 orientations. Larger counts provide marginal gains (+0.1–0.15pp) but increase parameters without improving interpretability.

J Training Dynamics

J.1 Convergence Analysis

We analyze the training dynamics of PHONSSM and its components.

The orthogonality loss decreases steadily, indicating progressive factorization of the component subspaces. The gap between training and validation accuracy remains small ($\sim$6pp at convergence), suggesting good generalization.

J.2 Component Learning Dynamics

Individual phonological components converge at different rates:

- **Handshape**: Fastest convergence (stabilizes by epoch 40); handshape is the most visually distinctive component with clear finger configurations.
- **Location**: Moderate convergence (epoch 60); requires learning spatial relationships relative to body landmarks.

Table 13: Sensitivity to loss weights on WLASL100.

λ_{ortho}	λ_{div}	Top-1 (%)
0.01	0.01	86.42
0.05	0.01	87.63
0.1	**0.01**	**88.37**
0.2	0.01	87.91
0.5	0.01	86.18
0.1	0.001	87.82
0.1	0.005	88.15
0.1	**0.01**	**88.37**
0.1	0.05	87.54
0.1	0.1	85.93

Table 14: Learning rate sensitivity on WLASL100.

Learning Rate	Top-1 (%)	Convergence
1×10^{-4}	86.82	95 epochs
2×10^{-4}	87.91	78 epochs
$\mathbf{3 \times 10^{-4}}$	**88.37**	**65 epochs**
5×10^{-4}	87.63	52 epochs
1×10^{-3}	85.41	41 epochs

- **Movement**: Slowest convergence (epoch 80); movement patterns require temporal integration across multiple frames.
- **Orientation**: Fast convergence (epoch 45); palm orientation has relatively few categories (8 prototypes).

J.3 Loss Landscape Analysis

We analyze the loss landscape by computing the Hessian eigenspectrum at convergence. The top eigenvalues are: $\lambda_1 = 12.4$, $\lambda_2 = 8.7$, $\lambda_3 = 5.2$, with rapid decay thereafter. The condition number $\kappa = \lambda_{\max}/\lambda_{\min} \approx 10^3$ indicates a well-conditioned optimization landscape, explaining the stable training dynamics.

K Limitations

We discuss limitations of PhonSSM in detail to guide future research.

K.1 Methodological Limitations

Isolated sign assumption. PhonSSM processes signs in isolation, assuming clean segmentation. Continuous signing involves co-articulation effects where adjacent signs influence each other's production, sign boundaries are ambiguous, and prosodic structure spans multiple signs. Extending to continuous recognition requires explicit segmentation or sequence-to-sequence modeling.

Fixed phonological structure. We adopt the classical Stokoe-Battison four-parameter model (handshape, location, movement, orientation). Alternative linguistic analyses propose:

- Autosegmental phonology with separate tiers for manual and non-manual features
- Prosodic structure including syllables and metrical feet
- Feature geometry with hierarchical organization of sub-components

Learned decompositions might discover more effective factorizations than hand-specified parameters.

Static prototypes. Component prototypes are fixed after training. Dynamic or instance-adaptive prototypes could better handle signer variation and novel phonological realizations.

Table 15: Effect of prototype counts on WLASL100.

N_h	N_l	N_m	N_o	Top-1 (%)
15	8	5	4	86.12
20	10	8	6	87.45
30	**15**	**10**	**8**	**88.37**
40	20	15	10	88.52
50	25	20	12	88.48

Table 16: Training convergence metrics on WLASL100.

Metric	Epoch 25	Epoch 50	Epoch 100
Train Loss	2.14	0.89	0.42
Val Loss	2.31	1.12	0.68
Train Acc (%)	52.3	78.6	94.2
Val Acc (%)	48.1	74.2	88.4
$\mathcal{L}_{\text{ortho}}$	0.42	0.18	0.08
$\mathcal{L}_{\text{div}}$	0.31	0.12	0.05

K.2 Evaluation Limitations

ASL-centric evaluation. All experiments use American Sign Language. While phonological principles (simultaneity, minimal pairs, compositional structure) are cross-linguistic, specific inventories differ:

- British Sign Language (BSL) uses different handshape inventory
- Chinese Sign Language has location contrasts not present in ASL
- Some sign languages distinguish two-handed vs. one-handed more strictly

Prototype counts and architectural choices may require adaptation for other sign languages.

Dataset biases. Training data comes primarily from controlled recording settings with:

- Adult native/fluent signers (under-representing learners, children, elderly)
- Neutral backgrounds and good lighting
- Citation-form signs (isolated, careful production)

K.3 Deployment Limitations

Pose estimation dependency. PhonSSM assumes high-quality pose landmarks from MediaPipe. Real-world degradation includes:

- Occlusion (self-occlusion, objects, other people)
- Motion blur during rapid movements
- Challenging lighting (backlighting, low light)
- Camera angles different from training distribution

Signer variation. Models may not generalize to:

- Signers with motor differences affecting articulation
- Regional/dialectal variation in sign production
- Non-native signers with L1 transfer effects

L Future Directions

Continuous sign language recognition. Extending PhonSSM to continuous signing requires: (1) implicit or explicit segmentation, (2) handling co-articulation, and (3) modeling sentence-level prosody. The phonological decomposition could inform CTC-style losses with component-level intermediate representations.

Cross-linguistic transfer. The phonological factorization may enable transfer learning across sign languages. Shared handshape or movement prototypes could bootstrap recognition for low-resource sign languages.

Multi-modal fusion. Combining skeleton input with facial landmarks (for non-manual markers) and RGB features (for fine-grained handshape) could address current limitations while preserving efficiency.

Learned phonological structure. Replacing hand-specified component pathways with learned factorization (e.g., via neural architecture search or information-theoretic objectives) might discover more effective decompositions.

Real-time applications. With 260 samples/second throughput, PHONSSM supports real-time deployment. Future work includes: mobile optimization, streaming recognition, and integration with sign language translation systems.